Movie Ratings Analysis Case Study Report

1. Introduction

This case study examines a 20-movie dataset to understand audience ratings, genre-level performance, box-office outcomes, and the incidence of highly rated films. The analysis uses Pandas for data preparation and calculation, with Matplotlib code supplied to reproduce the charts. Results apply only to the provided sample and should not be generalized to the broader movie market.

2. Dataset Description

The dataset contains **20 movies** and five original fields:

Column	Description
`Movie_ID`	Unique movie identifier
`Genre`	Movie genre: Action, Comedy, Drama, or Thriller
`Duration_Minutes`	Runtime in minutes
`IMDB_Rating`	IMDb rating on the dataset's rating scale
`Box_Office_Millions`	Box-office revenue, in millions

Sample data

Movie_ID	Genre	Duration_Minutes	IMDB_Rating	Box_Office_Millions
4001	Thriller	168	6.8	77.3
4002	Thriller	155	6.1	132.1
4003	Action	125	7.8	276.1

4004 Comedy	177	9.5	437.8
4005 Action	98	6.2	222.4

Setup code

```
import pandas as pd
import matplotlib.pyplot as plt

movies = pd.read_parquet("data/movie_ratings_dataset_csv.parquet")
```

3. Average IMDb Rating

The overall average IMDb rating is **7.21** (unrounded calculation: 7.2100).

```
average_rating = movies["IMDB_Rating"].mean()
print(f"Average IMDb rating: {average_rating:.2f}")
```

Interpretation: The sample's typical movie received a moderately strong rating, slightly above 7 out of 10.

4. Genre King

Average IMDb ratings were calculated by grouping movies by genre.

```
avg_rating_by_genre = (
    movies.groupby("Genre", as_index=False)
          .agg(Average_IMDB_Rating=("IMDB_Rating", "mean"),
               Movie_Count=("Movie_ID", "count"))
          .sort_values("Average_IMDB_Rating", ascending=False)
)

ax = avg_rating_by_genre.plot(
    kind="bar", x="Genre", y="Average_IMDB_Rating", legend=False,
    color="#356AA0", figsize=(8, 5)
)
ax.set_title("Average IMDb Rating by Genre")
ax.set_xlabel("Genre")
ax.set_ylabel("Average IMDb Rating")
plt.tight_layout()
plt.show()
```

Genre	Average IMDb Rating	Movie Count
Thriller	7.53	7

Action	7.50	5
Comedy	7.23	3
Drama	6.46	5

Genre King: Thriller, with an average IMDb rating of **7.53**. Thriller narrowly exceeds Action by about 0.03 rating points.

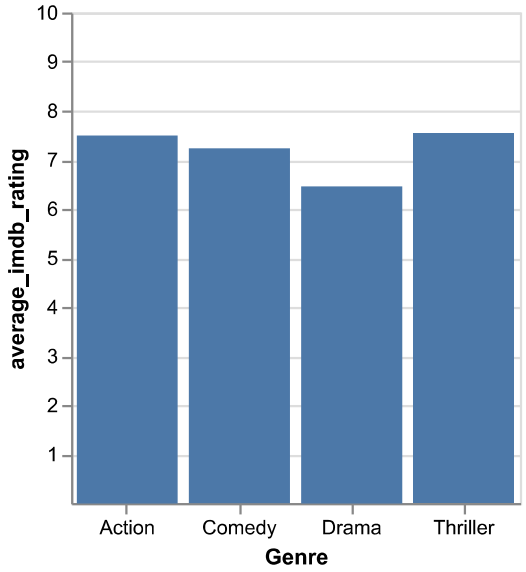


Figure 1. Average IMDb rating by genre. The y-axis is average IMDb rating; the x-axis is genre.

5. Financial Impact Analysis

Total box-office revenue was summed by genre. Values are in **millions**.

```
revenue_by_genre = (
    movies.groupby("Genre", as_index=False)
          .agg(Total_Box_Office_Millions=("Box_Office_Millions", "sum"),
               Movie_Count=("Movie_ID", "count"))
          .sort_values("Total_Box_Office_Millions", ascending=False)
)

ax = revenue_by_genre.plot(
    kind="bar", x="Genre", y="Total_Box_Office_Millions", legend=False,
    color="#2E8B57", figsize=(8, 5)
)
ax.set_title("Total Box Office Revenue by Genre")
ax.set_xlabel("Genre")
ax.set_ylabel("Total Box Office Revenue (Millions)")
plt.tight_layout()
plt.show()
```

Genre	Total Box Office Revenue (Millions)	Movie Count
Thriller	1,777.2	7
Action	1,383.8	5
Drama	1,339.9	5
Comedy	1,076.1	3

Highest-revenue genre: Thriller, totaling **$1,777.2 million** in this sample.

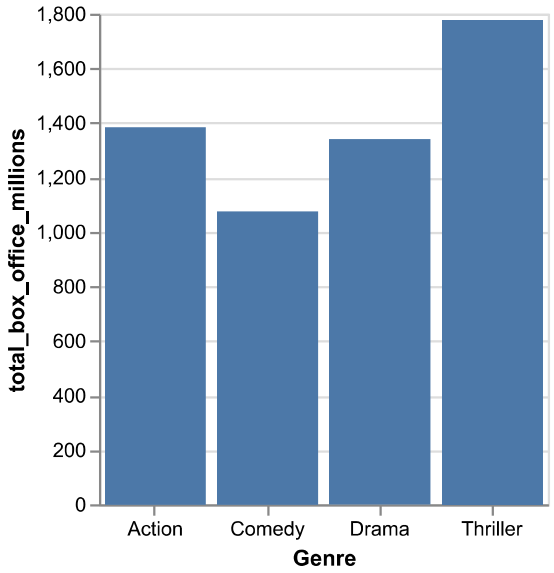


Figure 2. Total box-office revenue by genre. The y-axis is revenue in millions; the x-axis is genre.

Interpretation: Thriller leads both the average-rating and total-revenue comparisons in this dataset. However, the totals also reflect that Thriller has the largest number of movies (7), so they are not a per-film profitability measure.

6. Rating Distribution

```
plt.figure(figsize=(8, 5))
plt.hist(movies["IMDB_Rating"], bins=5, color="#7B68EE", edgecolor="black")
plt.title("Distribution of IMDb Ratings")
plt.xlabel("IMDb Rating")
plt.ylabel("Number of Movies")
plt.tight_layout()
plt.show()
```

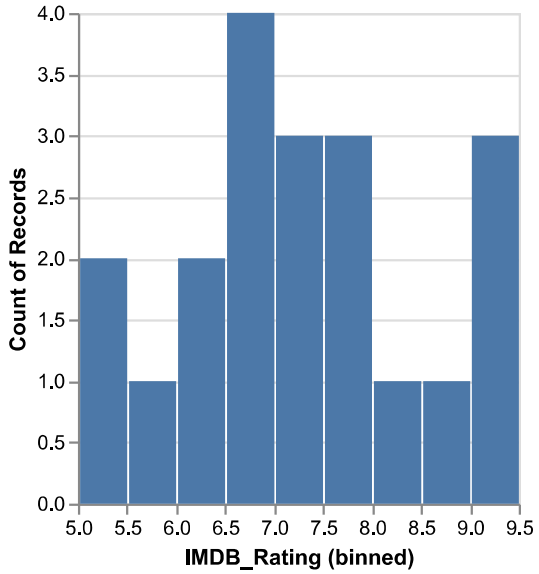


Figure 3. Distribution of IMDb ratings. The x-axis is IMDb rating and the y-axis is the number of movies.

Using one-point rating bands, the distribution is:

IMDb rating band	Movies
5.0–5.9	3
6.0–6.9	6
7.0–7.9	6
8.0–8.9	2
9.0–9.9	3

Interpretation: Ratings span from **5.2 to 9.5**. The largest concentrations are in the 6.0–6.9 and 7.0–7.9 bands (six films each), while five films meet or exceed the 8.0 hit threshold.

7. Hit Identification

A movie is labeled a **Hit** when `IMDB_Rating >= 8`.

```
movies["Hit"] = movies["IMDB_Rating"] >= 8
print(movies)
```

Updated dataset

Movie_ID	Genre	Duration_Minutes	IMDB_Rating	Box_Office_Millions	Hit
4001	Thriller	168	6.8	77.3	False
4002	Thriller	155	6.1	132.1	False
4003	Action	125	7.8	276.1	False
4004	Comedy	177	9.5	437.8	True
4005	Action	98	6.2	222.4	False
4006	Action	109	9.1	289.3	True
4007	Drama	154	6.6	440.9	False
4008	Action	95	7.3	136.5	False
4009	Thriller	150	7.6	397.3	False
4010	Thriller	104	6.6	486.9	False
4011	Thriller	131	8.6	370.3	True
4012	Drama	105	7.6	74.4	False
4013	Thriller	119	8.0	277.4	True
4014	Thriller	166	9.0	35.9	True
4015	Drama	166	5.2	277.1	False

4016	Comedy	140	5.3	343.2	False
4017	Action	153	7.1	459.5	False
4018	Comedy	176	6.9	295.1	False
4019	Drama	152	7.2	305.8	False
4020	Drama	99	5.7	241.7	False

8. Hit Movie Tally

```
hit_movies = movies.loc[movies["Hit"]].copy()
hit_count = movies["Hit"].sum()
hit_percentage = movies["Hit"].mean() * 100

print(f"Hit movies: {hit_count}")
print(f"Hit percentage: {hit_percentage:.1f}%")
print(hit_movies)
```

There are **5 Hit movies**, representing **25.0%** of the 20-movie sample.

Movie_ID	Genre	Duration_Minutes	IMDB_Rating	Box_Office_Millions
4004	Comedy	177	9.5	437.8
4006	Action	109	9.1	289.3
4011	Thriller	131	8.6	370.3
4013	Thriller	119	8.0	277.4
4014	Thriller	166	9.0	35.9

Interpretation: Thriller accounts for three of the five hits, Action and Comedy account for one each, and no Drama movie reaches the defined hit threshold in this sample.

9. Key Findings

- The overall average IMDb rating is **7.21**.
- **Thriller** is the top-rated genre (average **7.53**) and the highest-revenue genre (**$1,777.2 million** total).
- **Action** is nearly tied with Thriller on average rating (7.50), but trails in total sample revenue.
- Ratings are concentrated in the 6.0–7.9 range: **12 of 20 movies (60%)** fall in those two bands.
- **5 of 20 movies (25.0%)** meet the Hit definition of IMDb rating at least 8.0.
- The highest-rated movie is Movie_ID **4004** (Comedy, rating **9.5**), while the largest individual box-office figure is Movie_ID **4010** (Thriller, **$486.9 million**)—illustrating that the top rating and top revenue do not belong to the same movie.

10. Conclusion

The provided sample indicates that Thriller is the strongest all-around genre under the selected measures: it has the highest average IMDb rating, the greatest total box-office revenue, and the most hit movies. Still, the results are descriptive rather than causal. Revenue totals depend partly on the number of movies per genre, and the small 20-movie sample does not establish broader market patterns. Future analysis would benefit from more movies and additional fields—such as release year, production budget, marketing spend, and audience size—to evaluate profitability, trends, and the drivers of successful films more rigorously.